Quantum Mechanics I — Exercise Sheet 1

The qubit

Variant: Questions only

Conventions, as in Lecture 1: the inner product is antilinear in the first slot; $\psi(\theta,\phi) = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle$; the Bloch vector has components $r_a = \langle\psi,\sigma_a\psi\rangle$; a measurement along a unit vector $\mathbf{m}$ reads ±1 with probability $\frac{1}{2}(1 \pm \mathbf{m}\cdot\mathbf{r})$ and leaves the state at $\pm\mathbf{m}$; the Hamiltonian $H = c\mathbb{1} + \frac{\hbar\nu}{2}\mathbf{n}\cdot\boldsymbol{\sigma}$ turns the sphere about $\mathbf{n}$ at the rate ν.

Exercise 1 (rays and Bloch vectors). (i) Which of the vectors $\frac{1}{\sqrt{2}}(i\,|0\rangle - |1\rangle)$, $\frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$ and $\frac{1}{\sqrt{2}}(-|0\rangle - i\,|1\rangle)$ describe the same state as $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$? (ii) Let $\psi = \frac{1}{\sqrt{3}}\big((1+i)\,|0\rangle + |1\rangle\big)$. Check that ψ is a unit vector, find its Bloch vector, and find the angles θ, ϕ with $[\psi] = [\psi(\theta,\phi)]$. (iii) Give a unit vector for the state orthogonal to ψ, and its Bloch vector.

Exercise 2 (three axes determine a state). (i) Show that in the state with Bloch vector $\mathbf{r}$, $r_a = 2\,\mathrm{Prob}(+1 \text{ along } \mathbf{e}_a) - 1$ for $a = x, y, z$. (ii) Many atoms are prepared alike, and the measured fractions of $+1$ along x, y, z are 0.9, 0.5, 0.8. Find the state. (iii) What can one say if all three fractions are $\frac{1}{2}$? (iv) Show that the readings along x and z alone do not determine the state in general.

Exercise 3 (three magnets). Atoms prepared at $+\mathbf{e}_z$ pass a magnet along $\mathbf{m} = (\sin\theta, 0, \cos\theta)$, $0 \le \theta \le \pi$, and then a magnet along z. Both readings are kept, and nothing acts between the magnets. (i) Find the probabilities of the four pairs of readings. (ii) Find the probability that the second magnet reads -1. For which θ is it largest? What is it with the middle magnet removed? (iii) ⋆ Replace the middle magnet by N magnets along $\mathbf{m}_k = (\sin(k\pi/N), 0, \cos(k\pi/N))$, $k = 1, \dots, N$, so that the last one points along $-\mathbf{e}_z$. Find the probability that all N read $+1$, its value for $N = 1$, 2 and 10, and its limit as $N \to \infty$.

Exercise 4 (after one turn). Let $H = c\mathbb{1} + \frac{\hbar\nu}{2}\mathbf{n}\cdot\boldsymbol{\sigma}$ with $\mathbf{n}$ a unit vector and $\nu > 0$. (i) Show that $U_t = e^{-itH/\hbar} = e^{-ict/\hbar}\big(\cos(\nu t/2)\mathbb{1} - i\sin(\nu t/2)\,\mathbf{n}\cdot\boldsymbol{\sigma}\big)$. (ii) Show that $U_{2\pi/\nu} = -e^{-2\pi i c/(\hbar\nu)}\mathbb{1}$. After one turn of the sphere, has the state come back? When has the vector? (iii) Let $c = 0$. Find the least $T > 0$ with $U_T = \mathbb{1}$. (iv) Show that U_t multiplies the state at $+\mathbf{n}$ by $e^{-iE_+t/\hbar}$, where $E_+ = c + \frac{1}{2}\hbar\nu$.

Exercise 5 (the sphere as a phase space, for every axis). In classical mechanics the state of a system is a point (q,p) of its *phase space*, and the energy, a function $h(q,p)$, moves the points by *Hamilton's equations* $\dot q = \partial h/\partial p$ and $\dot p = -\partial h/\partial q$. We compare this with the motion on the Bloch sphere.

Let $\mathbf{n}$ be a unit vector, and choose unit vectors $\mathbf{u}$, $\mathbf{v}$ with $\mathbf{u}\times\mathbf{v} = \mathbf{n}$. Away from $\pm\mathbf{n}$, let q be the angle of the point $(\mathbf{u}\cdot\mathbf{r}, \mathbf{v}\cdot\mathbf{r})$ of the plane, and $p = \frac{\hbar}{2}\,\mathbf{n}\cdot\mathbf{r}$. Show that the precession of Theorem 1.12 reads $\dot q = \nu$, $\dot p = 0$, and that these are Hamilton's equations for $h = c + \frac{\hbar\nu}{2}\,\mathbf{n}\cdot\mathbf{r}$, the mean energy in the state. So for the motion the sphere is a phase space. For the readings it is not (§ 1.7).

Exercise 6 (Larmor's sense of rotation). An electron has the gyromagnetic ratio $\gamma_e = -1.761 \cdot 10^{11}\ \mathrm{s^{-1}\,T^{-1}}$, a proton $\gamma_p = 2\pi \cdot 42.577\ \mathrm{MHz\,T^{-1}}$. Each sits in the field $\mathbf{B} = B\,\mathbf{e}_z$, $B = 1$ mT, with $H = -\gamma \frac{\hbar}{2} \boldsymbol{\sigma} \cdot \mathbf{B}$. For each, find $\mathbf{n}$, the frequency $\nu/2\pi$, the period $2\pi/\nu$, and the sense in which the Bloch vector turns, seen from the tip of $\mathbf{e}_z$.

Exercise 7 ⋆. Show that every self-adjoint 2×2 matrix A with $A^2 = \mathbb{1}$ and $A \neq \pm\mathbb{1}$ is $\mathbf{m} \cdot \boldsymbol{\sigma}$ for a unit vector $\mathbf{m}$. What are the readings of such an observable?